EEE-301
Data and Telecommunication

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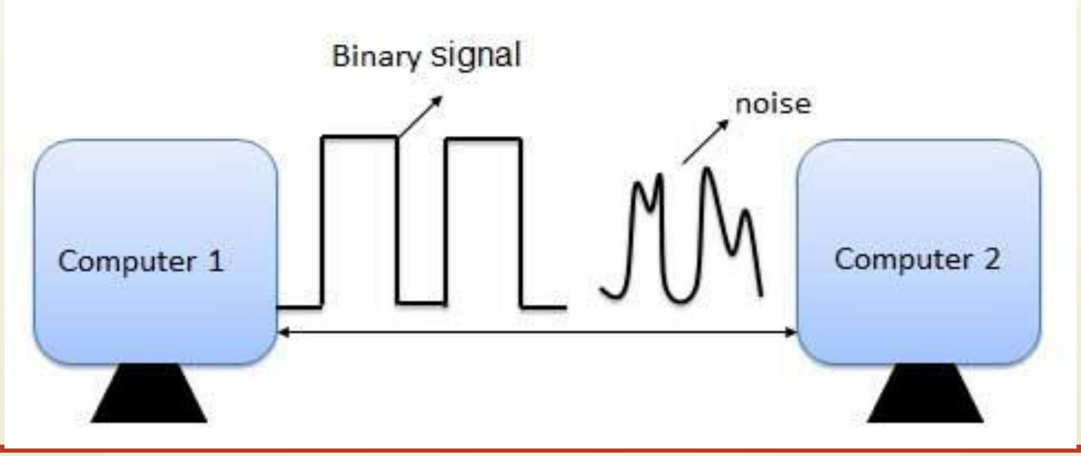
Chapter 10
Error Detection And Correction

Book reference: Data Communication and Networking
By- Behrouz A Forouzan

What Is Error?

- Error is a condition when the output information does not match with the input information. During transmission, digital signals suffer from noise that can introduce errors in the binary bits travelling from one system to other. That means a 0 bit may change to 1 or a 1 bit may change to 0.



Error Detection And Correction

- Data can be corrupted during transmission. For reliable communication, errors must be detected and corrected Error Types.

Types of Errors:

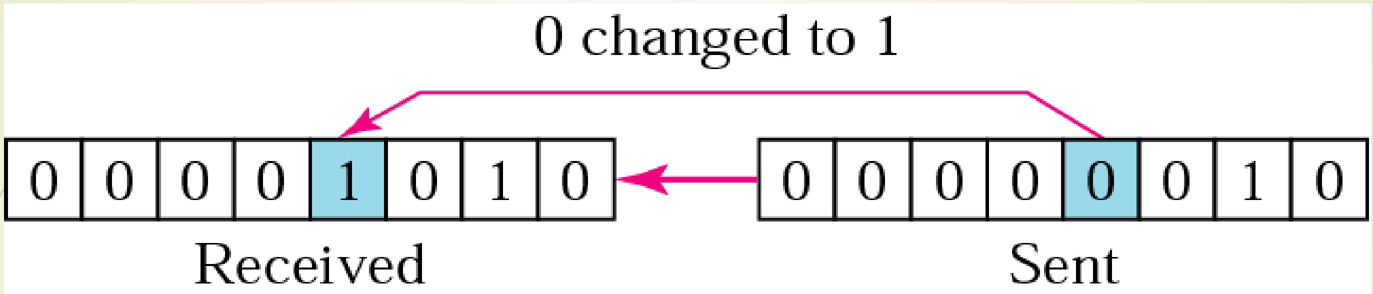
- Single-Bit Error

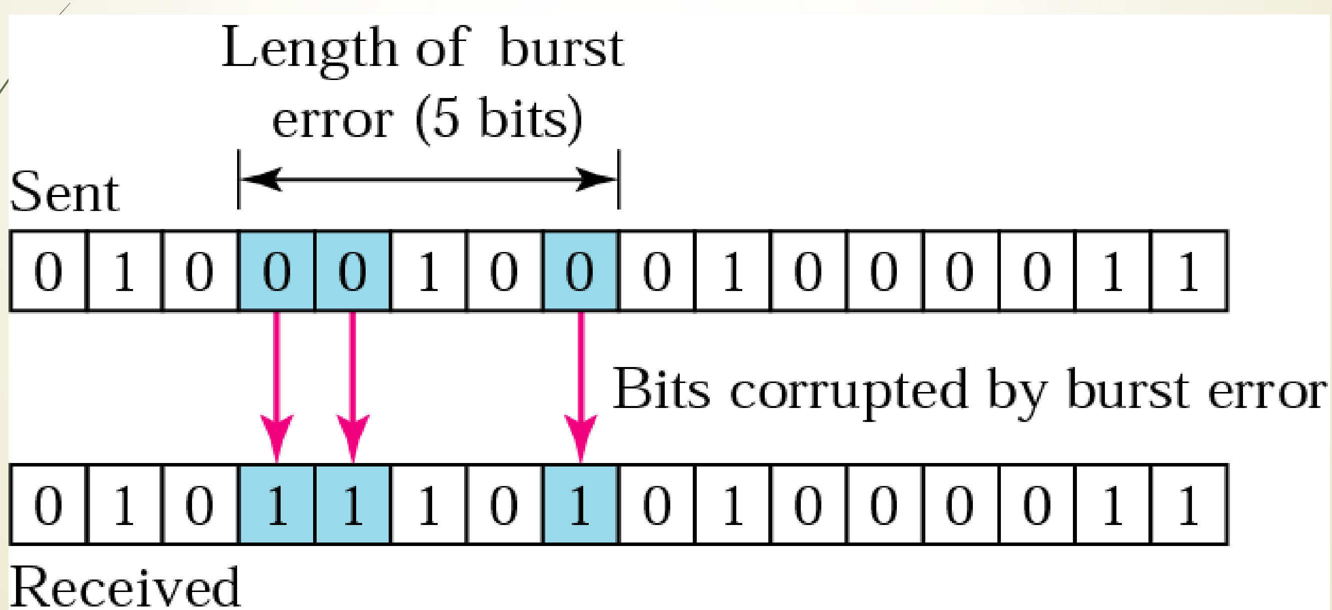
  – one bit in the data unit has changed

- Burst Error

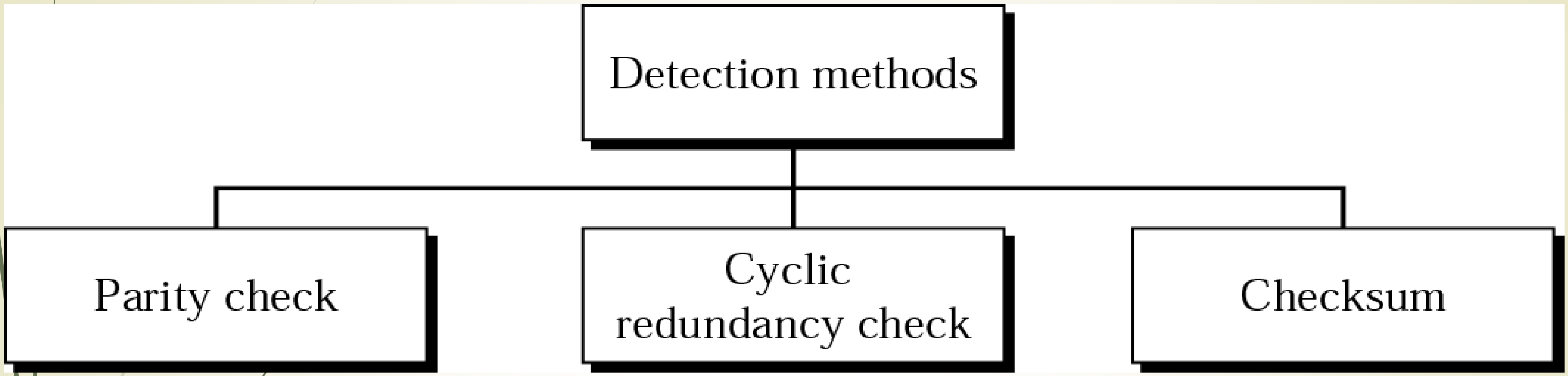
  – 2 or more bits in the data unit have changed

Single Bit Error Vs. Burst Error





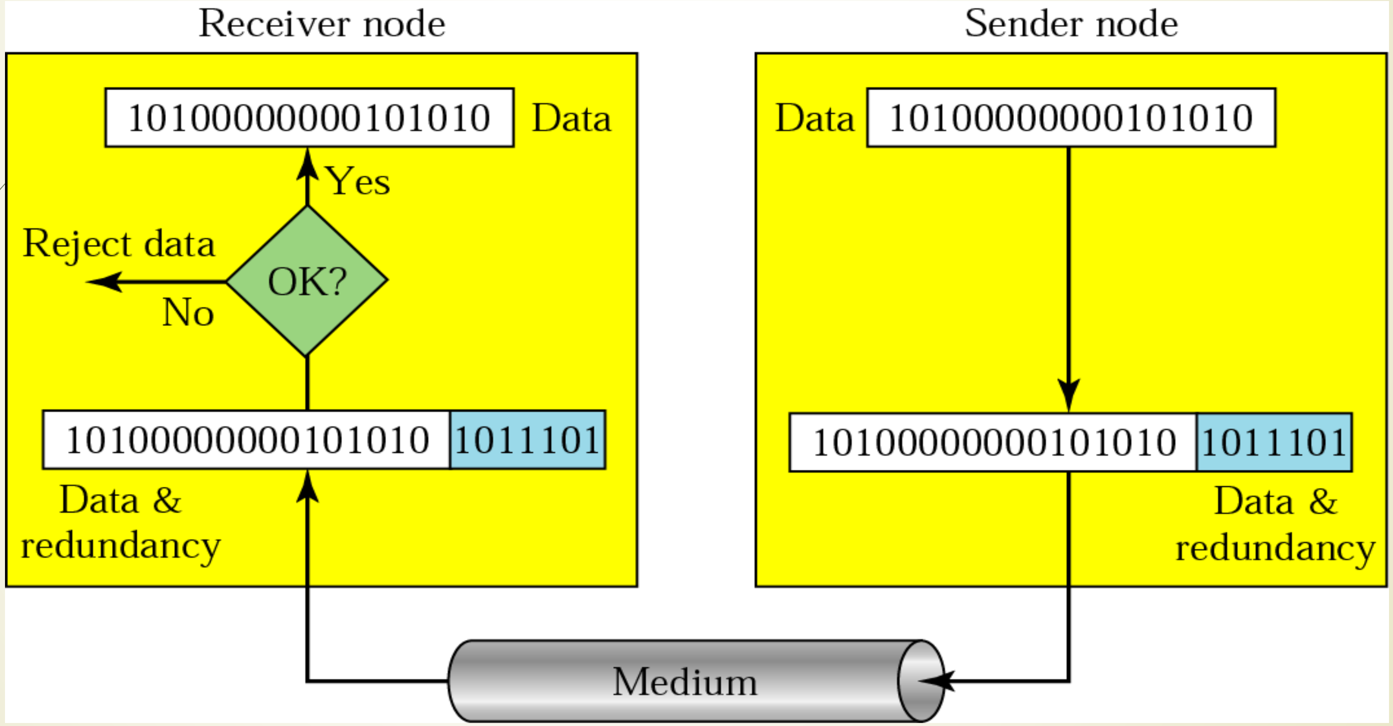
ERROR DETECTION



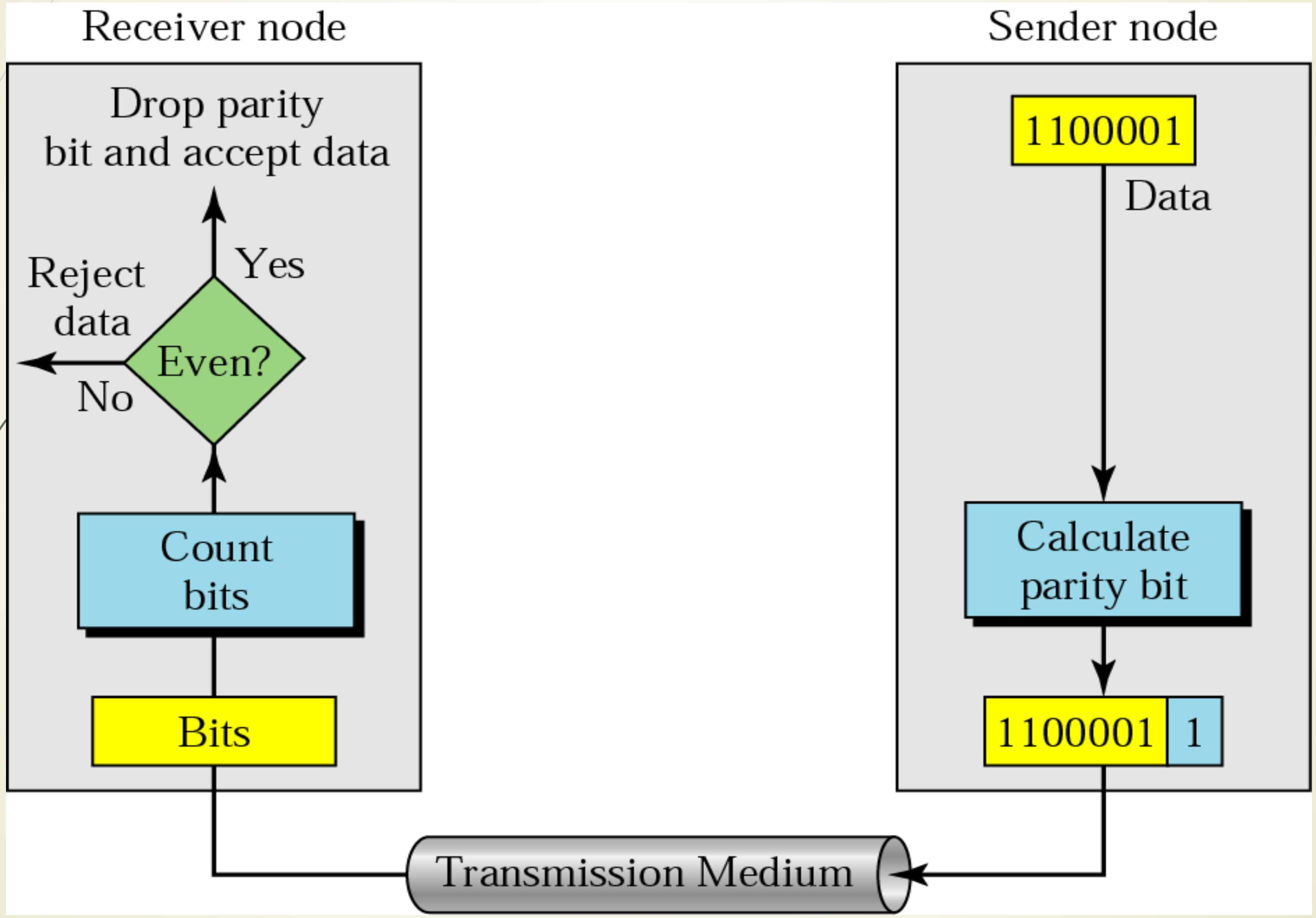
Error detection uses the concept of redundancy, which means adding extra bits for detecting errors at the destination.

Redundancy For Error Detection

Adding extra bits for error detection at the destination. Three types: Parity check, Cyclic redundancy check, Checksum.



Simple Parity Check

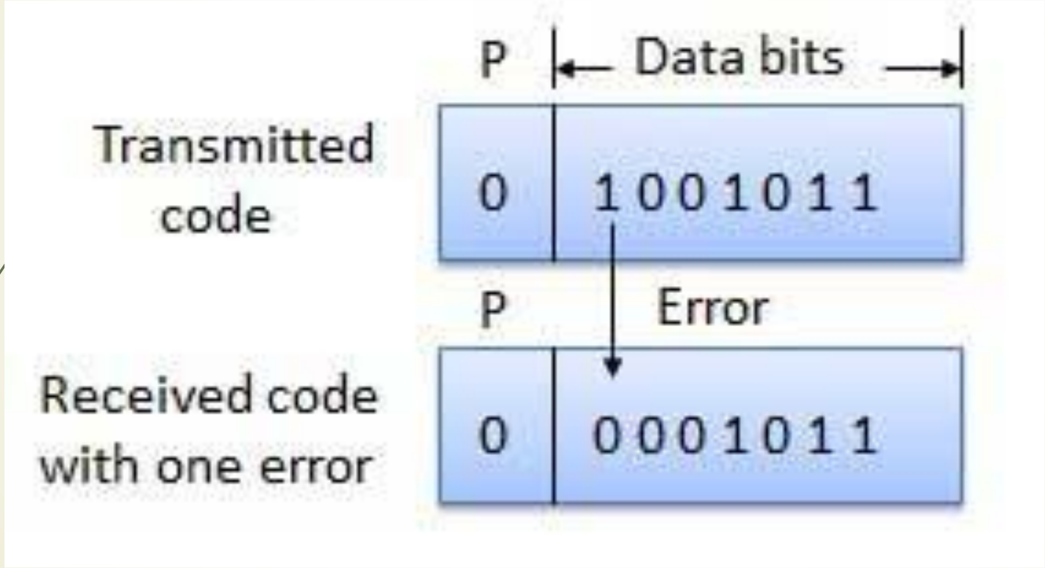


Simple Parity Check

- How Does Error Detection Take Place?
- Parity checking at the receiver can detect the presence of an error if the parity of the receiver signal is different from the expected parity. That means, if it is known that the parity of the transmitted signal is always going to be "even" and if the received signal has an odd parity, then the receiver can conclude that the received signal is not correct. If an error is detected, then the receiver will ignore the received byte and request for retransmission of the same byte to the transmitter.

Simple Parity Check

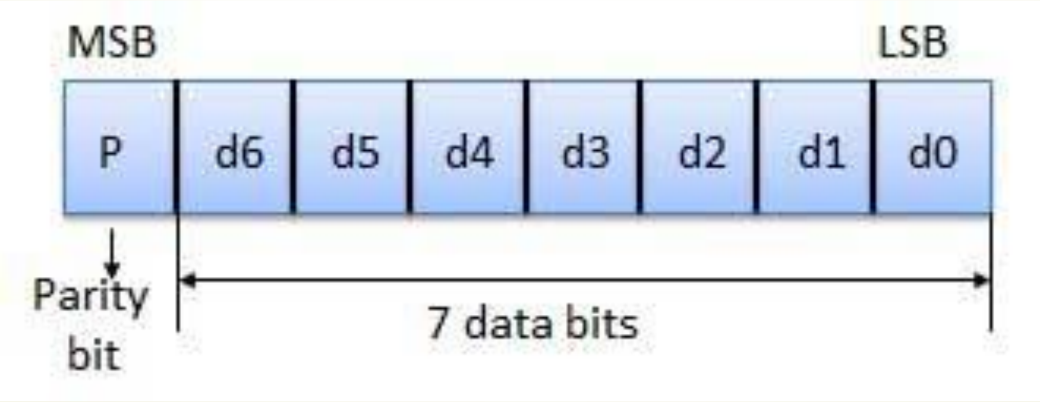
- How Does Error Detection Take Place?



Simple Parity Check

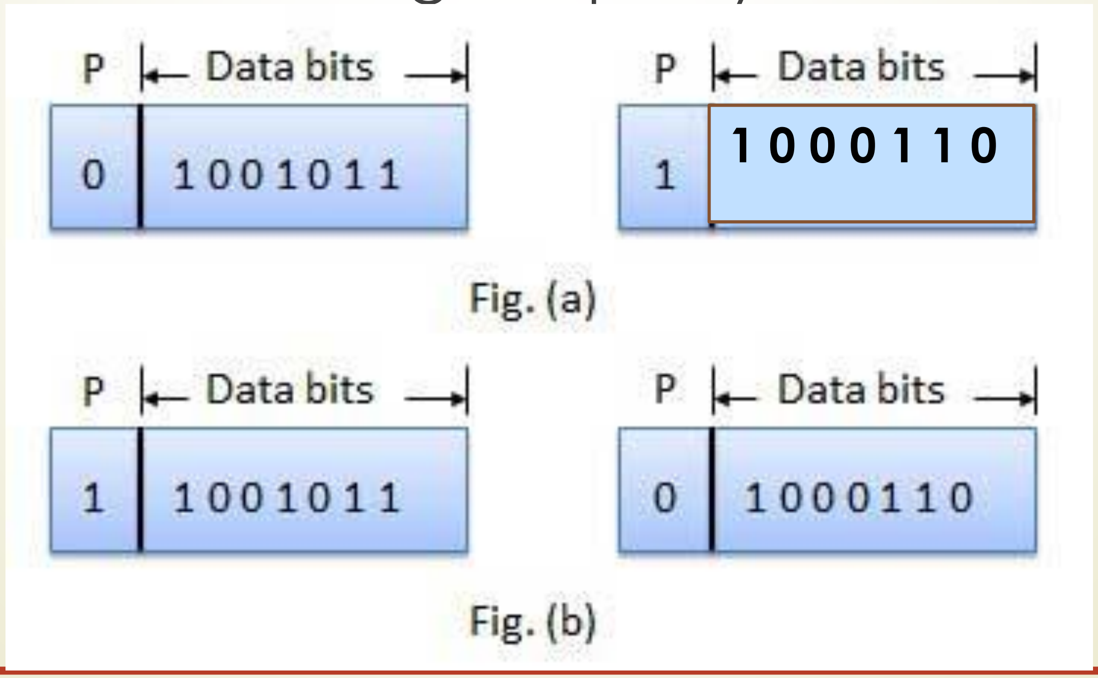
Parity Checking of Error Detection.

- It is the simplest technique for detecting and correcting errors. The MSB of an 8-bits word is used as the parity bit and the remaining 7 bits are used as data or message bits. The parity of 8-bits transmitted word can be either even parity or odd parity.

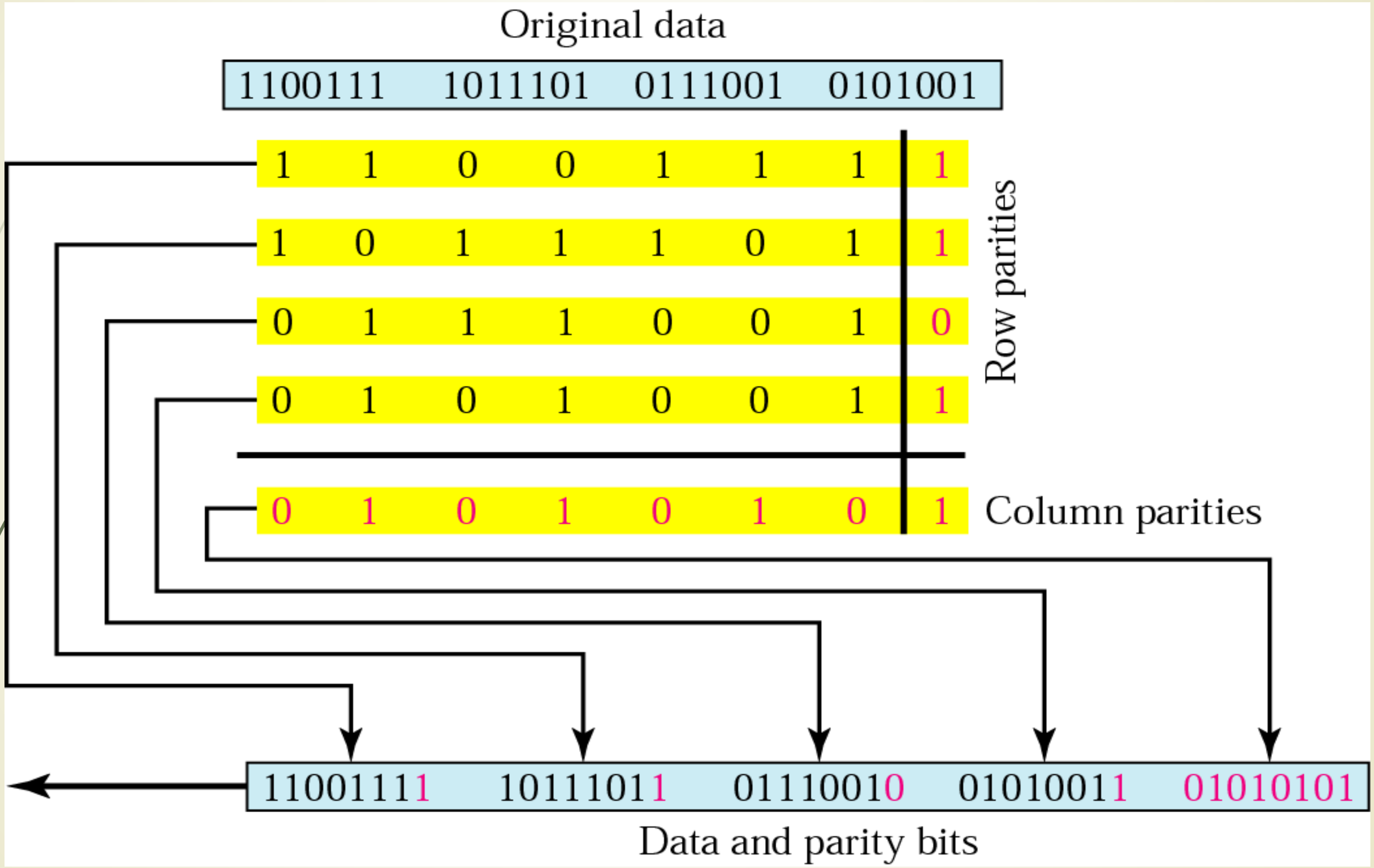


Parity Types

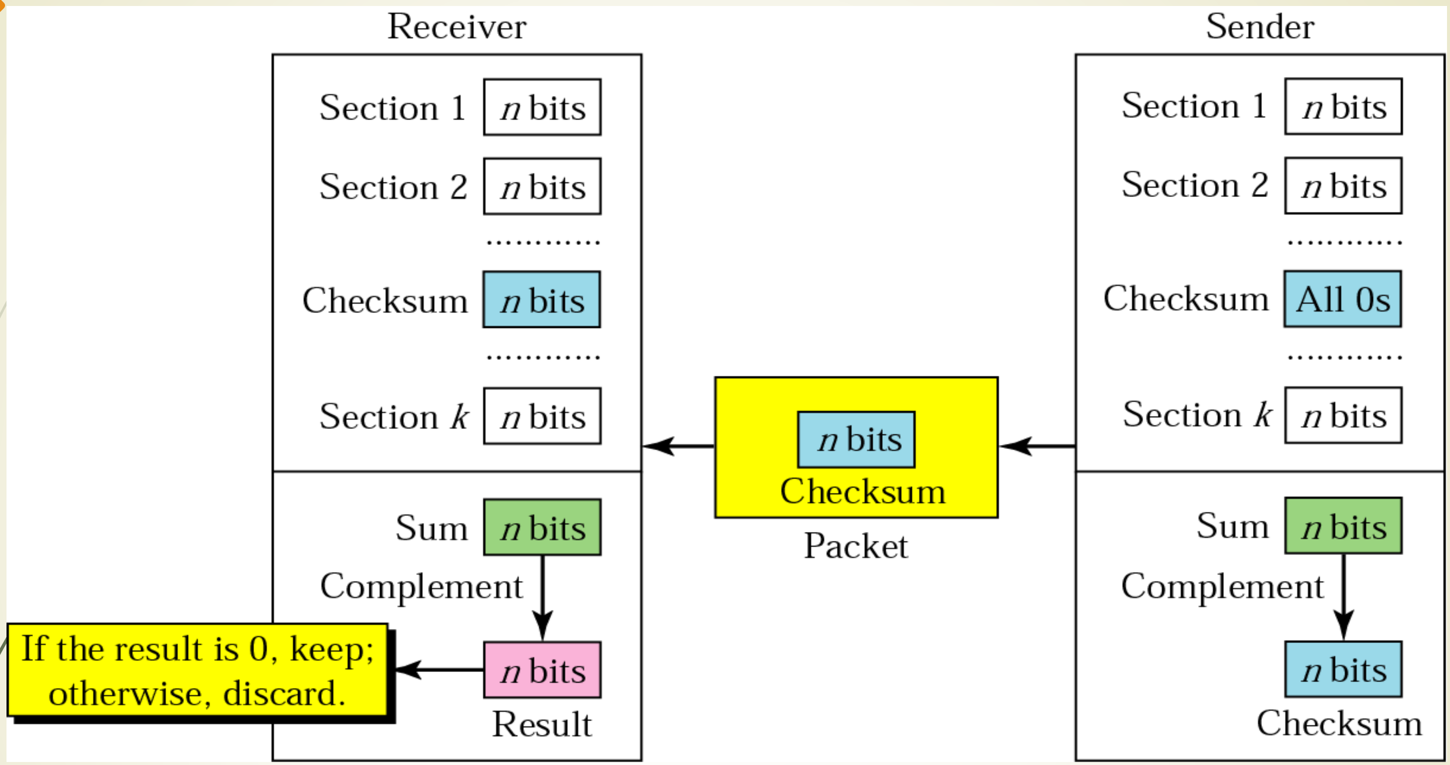
- **Even parity** -- Even parity means the number of 1's in the given word including the parity bit should be even (2,4,6,....).
- **Odd parity** -- Odd parity means the number of 1's in the given word including the parity bit should be odd (1,3,5,....).

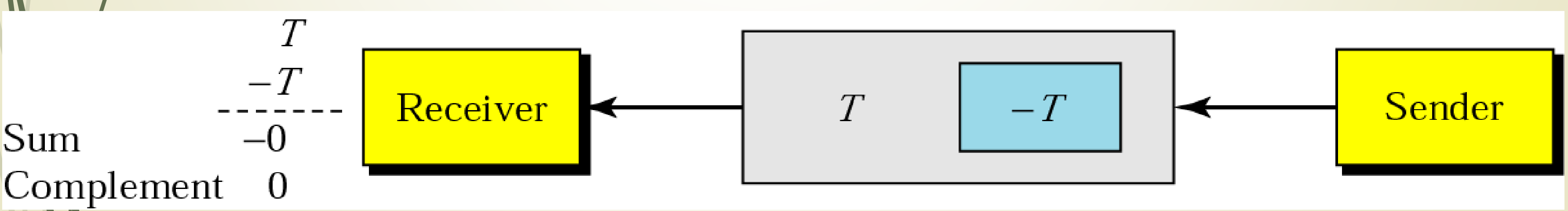


2-d Parity Check



Checksum


Receiver
Sender
Section 1 n bits
Section 2 n bits
Checksum n bits
Section k n bits
Checksum All 0s
n bits
Checksum
Packet
Sum n bits
Complement
n bits
Result
Checksum
If the result is 0, keep; otherwise, discard.


T
−T
Sum −0
Complement 0
Receiver
T
−T
Sender

Checksum: Sending

Suppose the following block of 16 bits is to be sent using a checksum of 8 bits.

10101001 00111001

The numbers are added using one's complement

	10101001
	00111001

Sum	11100010
Checksum	**00011101**

The pattern sent is 10101001 00111001 **00011101**

Checksum: Receiving

Now suppose the receiver receives the pattern sent in Example 7 and there is no error.

10101001 00111001 00011101

When the receiver adds the three sections, it will get all 1s, which, after complementing, is all 0s and shows that there is no error.

	10101001
	00111001
	00011101
Sum	11111111
Complement	**00000000** means that the pattern is OK.

Checksum: Error

A burst error of length 5 that affects 4 bits.

10101111 11111001 00011101

When the receiver adds the three sections, it gets

10101111

11111001

00011101

Partial Sum **1** 11000101

Carry **1**

Sum 11000110 =>

Complement **00111001!!!??**

CRC (Cyclic Redundancy Check)

- The cyclic redundancy check (CRC) is a technique used to detect errors in digital data.
- commonly used in digital telecommunications networks and storage devices such as hard disk drives.
- This technique was invented by W. Wesley Peterson in 1961 and further developed by the CCITT (**Consultative Committee for International Telephony and Telegraphy**).
- It is based on binary division and is also called polynomial code checksum.

CRC (Cyclic Redundancy Check)

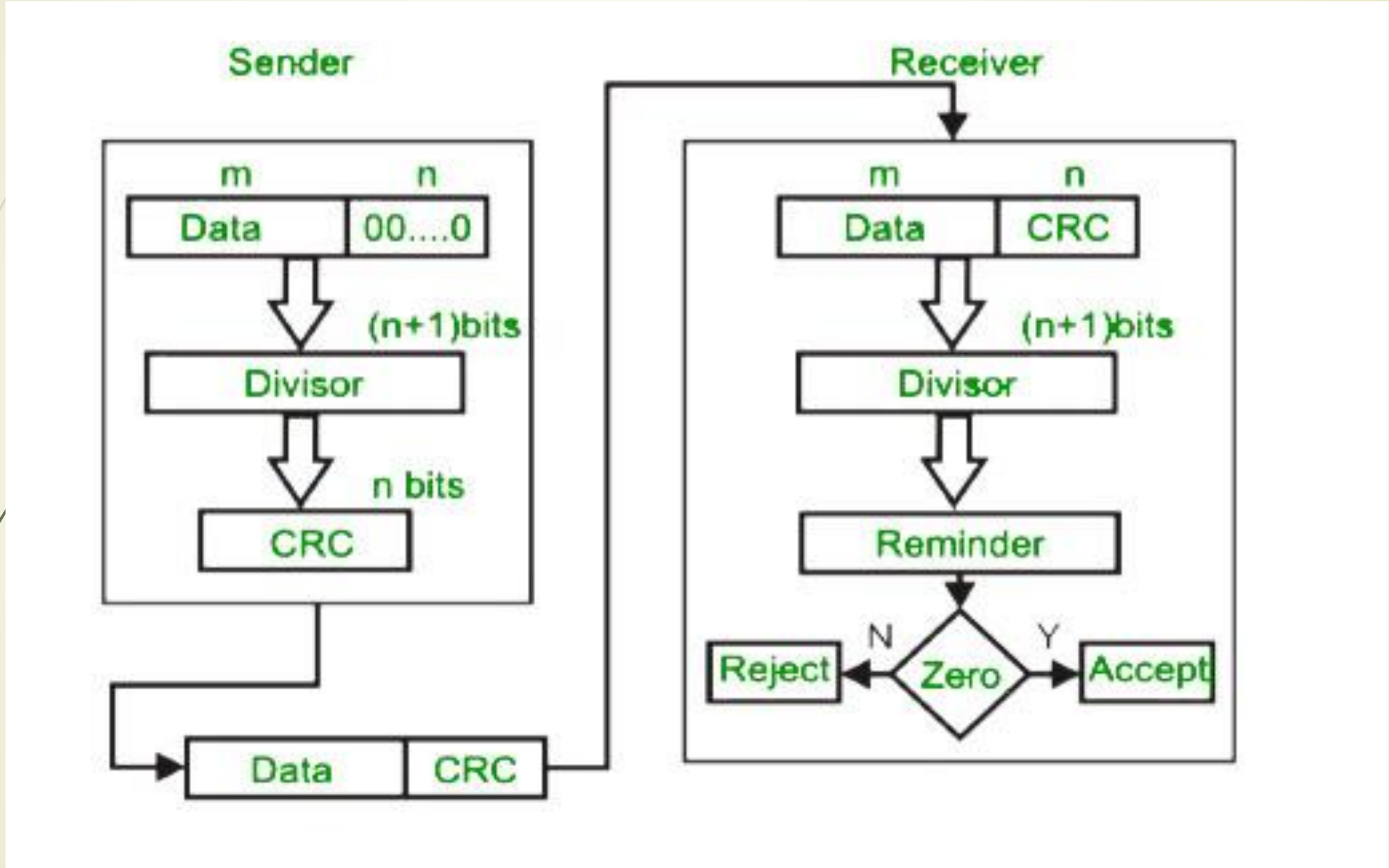


Figure: CRC generator and checker

CRC (Cyclic Redundancy Check)

Requirements of CRC :

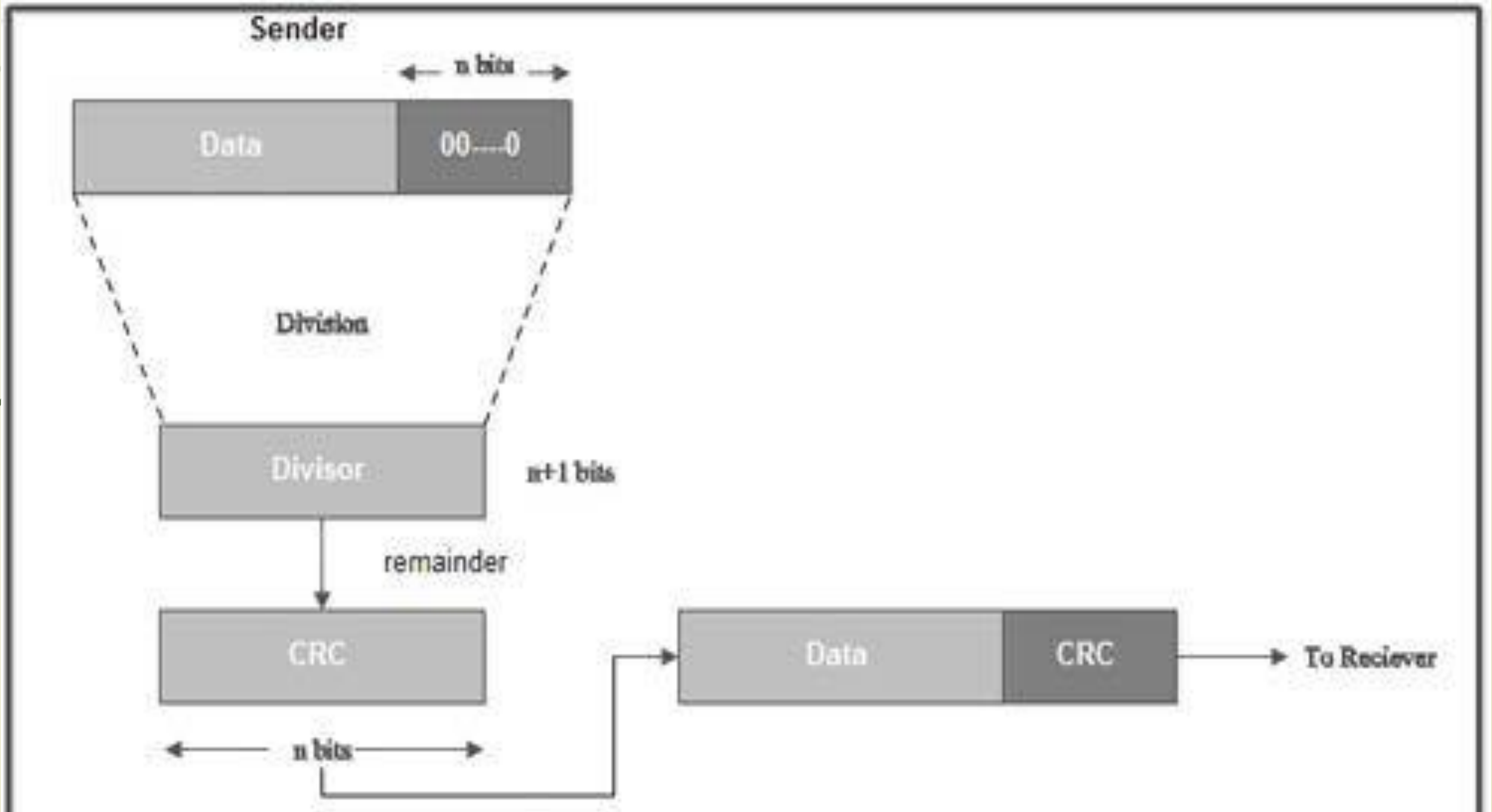
A CRC will be valid if and only if it satisfies the following requirements:

1. It should have exactly one less bit than divisor.
2. Appending the CRC to the end of the data unit should result in the bit sequence which is exactly divisible by the divisor.

CRC (Cyclic Redundancy Check)

The various steps followed in the CRC method are

1. A string of n as is appended to the data unit. The length of predetermined divisor is n+ 1.



CRC Generator at sender side

CRC (Cyclic Redundancy Check)

2. The newly formed data unit *i.e.* original data + string of n as are divided by the divisor using binary division and remainder is obtained. This remainder is called CRC.

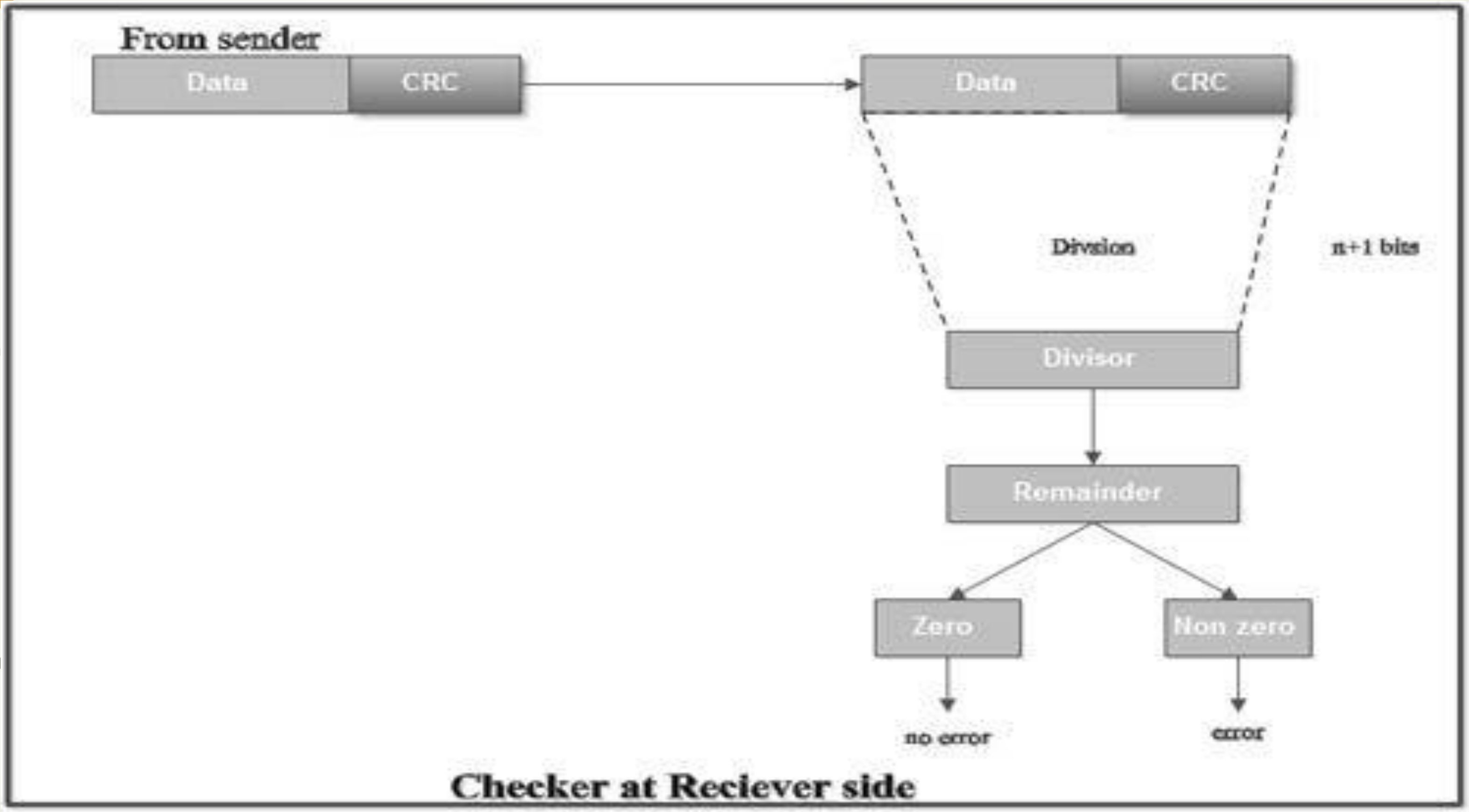
3. Now, string of n 0s appended to data unit is replaced by the CRC remainder (which is also of n bit).

4. The data unit + CRC is then transmitted to receiver.

5. The receiver on receiving it divides data unit + CRC by the same divisor & checks the remainder.

6. If the remainder of division is zero, receiver assumes that there is no error in data and it accepts it.

CRC (Cyclic Redundancy Check)

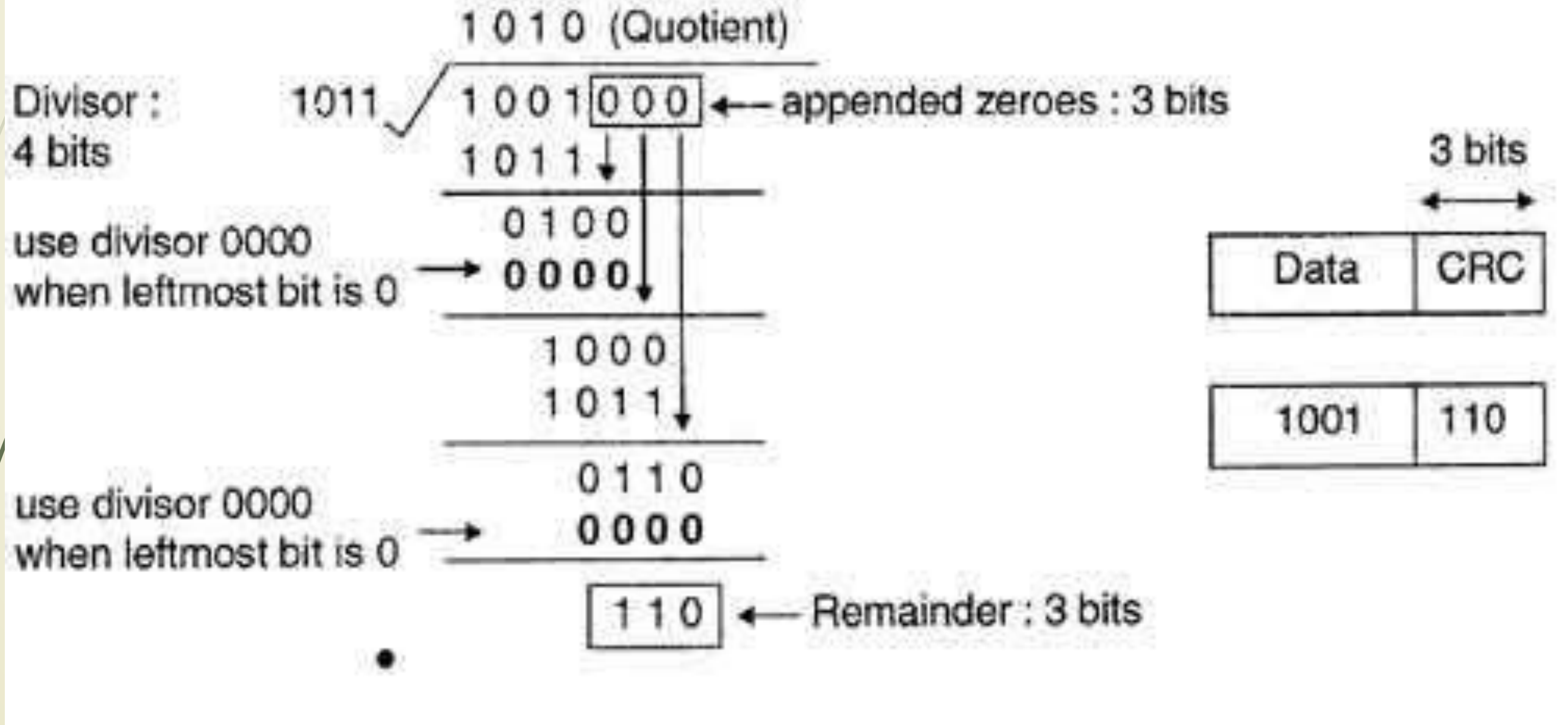


7. If remainder is non-zero then there is an error in data and receiver rejects it.

CRC (Cyclic Redundancy Check)

• For example, if data to be transmitted is 1001 and predetermined divisor is 1011. The procedure given below is used:

1. String of 3 zeroes is appended to 1011 as divisor is of 4 bits. Now newly formed data is 1011000.

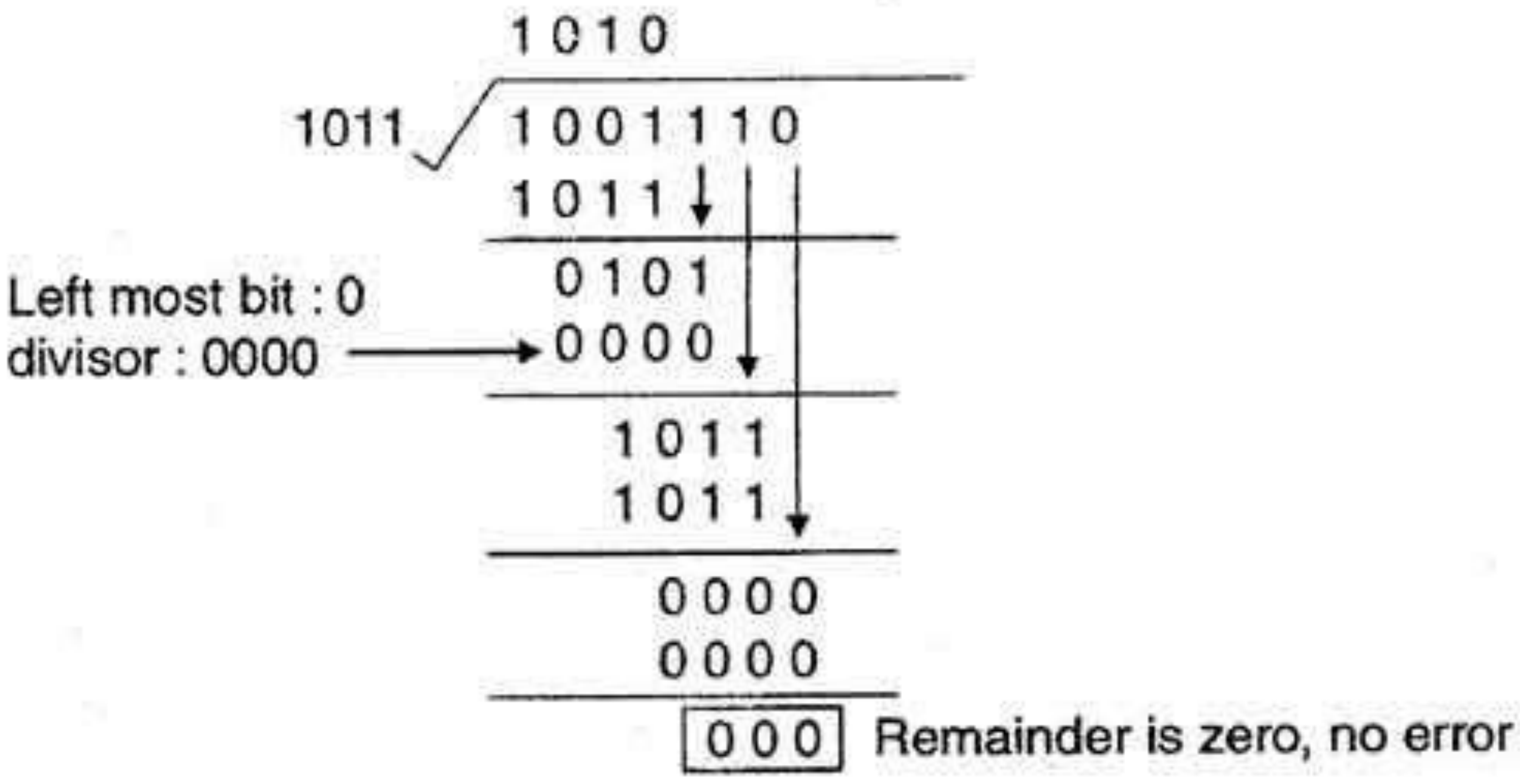


CRC generated (Binary division)

CRC (Cyclic Redundancy Check)

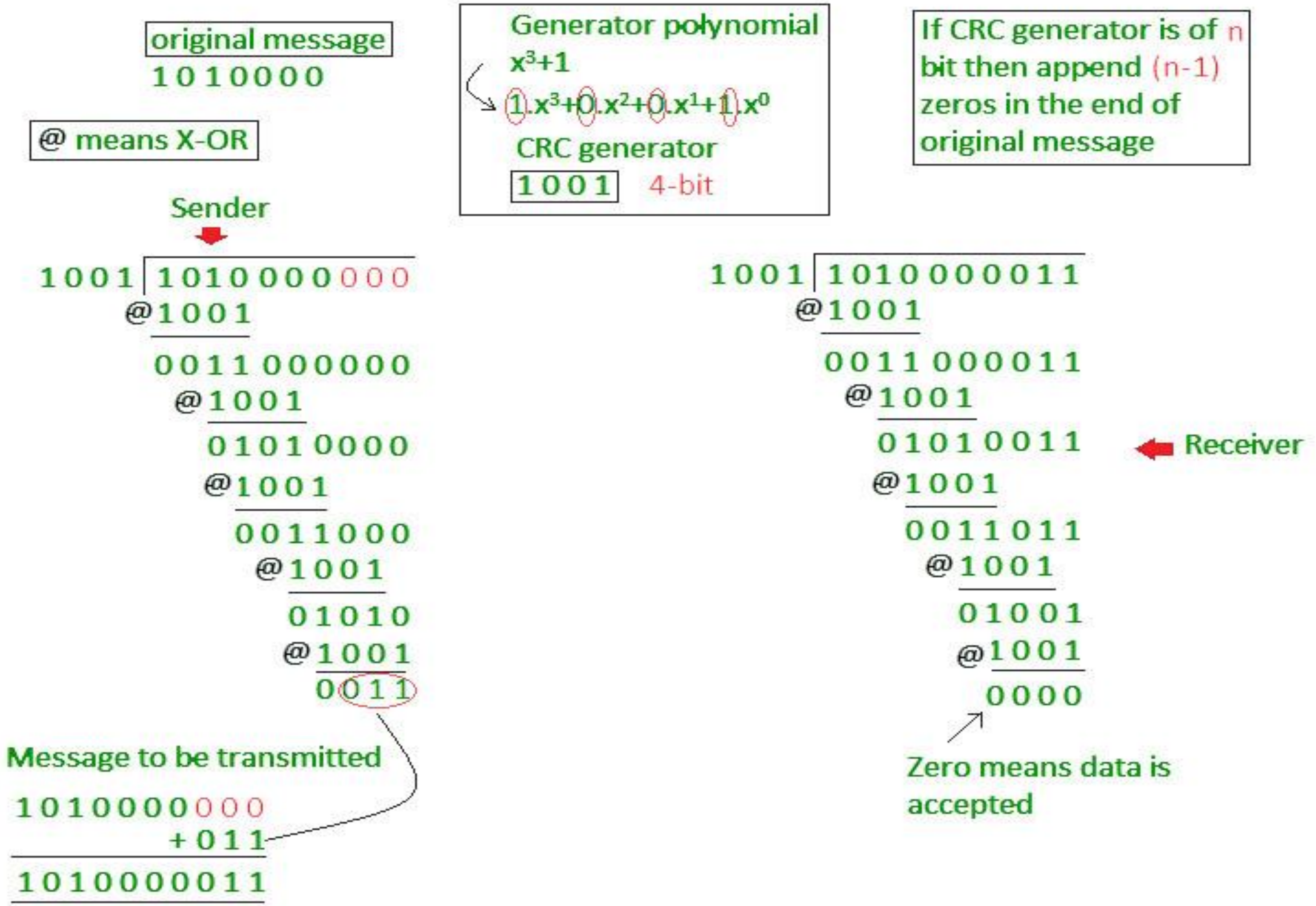
1. Data unit 1011000 is divided by 1011.
2. During this process of division, whenever the leftmost bit of dividend or remainder is 0, we use a string of 0s of same length as divisor. Thus in this case divisor 1011 is replaced by 0000.
3. At the receiver side, data received is 1001110.
4. This data is again divided by a divisor 1011.
5. The remainder obtained is 000; it means there is no error.

CRC (Cyclic Redundancy Check)



CRC decoded (binary division)

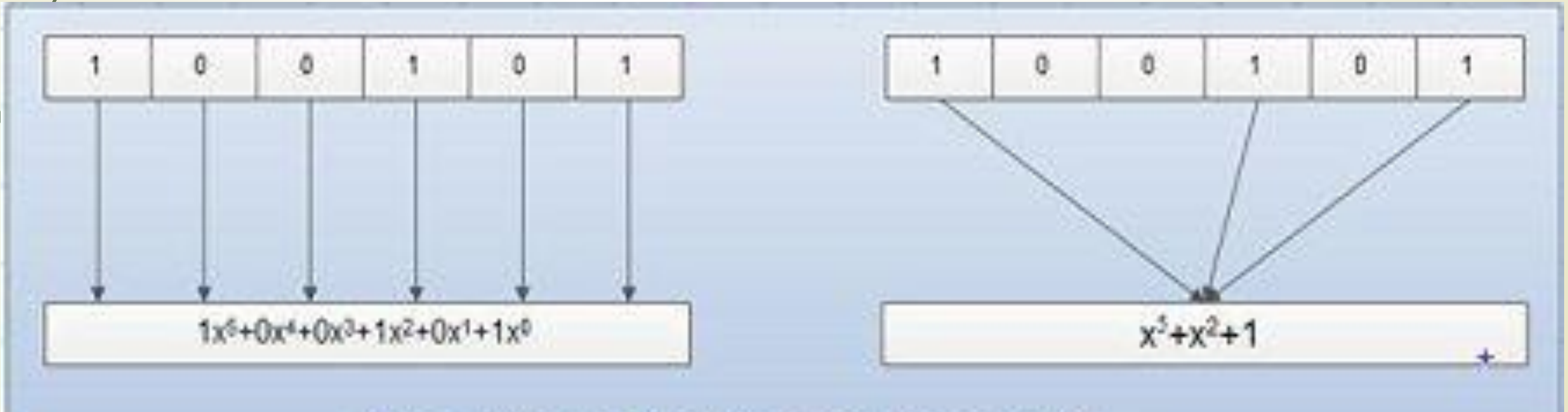
CRC (Cyclic Redundancy Check)



Another example of CRC

Polynomials

- A pattern of 0s and 1s can be represented as a polynomial with coefficient of 0 and 1.
- Here, the power of each term shows the position of the bit and the coefficient shows the values of the bit.
- For example, if binary pattern is 100101, its corresponding polynomial representation is $x^5 + x^2 + 1$. Figure shows the polynomial where all the terms with zero coefficient are removed and x J is replaced by x and XO by 1.



Binary pattern and its polynomial representation

Polynomials

- There are several different standard polynomials used by popular protocols for CRC generation.

These are:

- ATM header: Asynchronous Transmission Mode
- ATM ALL: **AAL** (*ATM Adaptation Layer*)
- HDLC: **HDLC** (High-level Data Link Control)

Name	Polynomial	Application
CRC-8	$x^8 + x^2 + x + 1$	ATM header
CRC-10	$x^{10} + x^9 + x^5 + x^4 + x^2 + 1$	ATM AAL
CRC-16	$x^{16} + x^{12} + x^5 + 1$	HDLC
CRC-32	$x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$	LANs

Error Correction

- **Two most common techniques:**

i. **Retransmission/Backward Error Correction**

ii. **Forward Error Correction**

Error Correction

- **Retransmission/Backward Error Correction:**
- When the receiver detects an error in the data received, it requests back the sender to retransmit the data unit.
- It is simple and can only be efficiently used where retransmitting is not expensive.
- For example, fiber optics.
- But in case of wireless transmission, retransmitting may cost too much.
- In this case, Forward Error Correction is used.

Forward Error Correction

- **Forward Error Correction (FEC)**

- When the receiver detects some error in the data received, it executes error-correcting code, which helps it to auto-recover and to correct some kinds of errors.
- To correct the error in data frame, the receiver must know exactly which bit in the frame is corrupted. To locate the bit in error, redundant bits are used as parity bits for error detection. For example, we take ASCII words (7 bits data), then there could be 8 kind of information we need: first seven bits to tell us which bit is error and one more bit to tell that there is no error.


Thanks